The Shape of Good Behavior: A Geometric Approach to AI Alignment

Michael Lee

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Abstract

Modern AI alignment methods like Reinforcement Learning from Human Feedback (RLHF) project rich, multi-dimensional human preferences onto a single scalar reward. This oversimplification leads to reward hacking, where models exploit gaps between the proxy and true objectives. We propose a geometric framework that treats preferences as living on a high-dimensional *Reward Manifold*. Using tools from differential topology—Hodge decomposition and sheaf cohomology—we separate consistent feedback (the gradient) from cyclic inconsistencies (the curl) to detect ambiguity or deception, and stitch together locally valid feedback patches into a globally coherent reward manifold.

1 From Social Choice to Differentiable Manifolds

1.1 Decoding Topological Features

Social choice theory tells us that aggregating preferences often leads to paradoxes (Condorcet cycles). In our framework, these paradoxes are not failures; they are topological features on the Reward Manifold, which can be decoded using Betti numbers (H_n).

- H_0 **(Connected Components):** This number represents the count of distinct, disconnected regions of valid, safe behavior. If $H_0 > 1$, the manifold is fragmented, suggesting multiple, self-contained local optima of alignment that cannot be navigated between without passing through unaligned or "bad" behavior.
- H_1 **(Cyclic Inconsistencies):** A non-vanishing 1-cohomology class ($H^1 \neq 0$) captures the rotational component (the Curl). It manifests as cyclic/intransitive preferences (e.g., Condorcet cycles), indicating inconsistencies or feedback loops in the preference data.
- H_2 **(Vulnerabilities and Voids):** This number represents the count of two-dimensional "voids" or "cavities" within the high-reward space. These could characterize the "black holes" or unrecoverable singularities—complex, hidden safety cliffs—that are entirely surrounded by aligned behavior but, once entered, cannot be exited.

Instead of discarding inconsistent data, we use these topological features to characterize the curvature of the reward space. We turn the discrete, noisy problem of social choice into a continuous, differentiable problem of manifold learning. This allows us to optimize policies that don't just "maximize reward," but "navigate the manifold," avoiding topological singularities (black holes) and following the gradient of the consistent component.

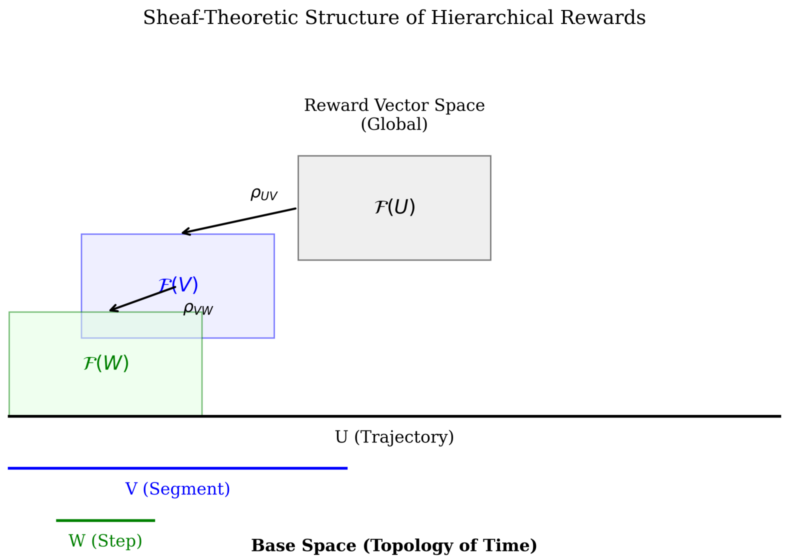

Figure 1: Sheaf structure over the reward manifold. Local sections (human feedback patches) must satisfy compatibility conditions to glue into global sections (consistent value functions).

2 The Storytelling Machine: A Concrete Paradigm

To validate this geometric intuition, we propose a shift in experimental design: treat alignment as a storytelling problem. The world is a book; each page is a scene description; the agent writes actions and the world writes consequences.

2.1 Text Adventure Games as Natural Language MDPs

Consider the classic text adventure:

```
> You are in a dark forest. Paths lead NORTH and EAST.
> A rusty lantern lies on the ground.
Your choice of actions is:
> TAKE LANTERN
You pick up the lantern.
> LIGHT LANTERN
The lantern flickers to life, revealing a hidden path to the WEST.
```

This simple interaction contains the complete structure of a Markov Decision Process:

- **State:** Natural language description of the world.
- **Action:** A verb-noun command that transforms state.
- **Transition:** Causal evolution according to world rules.
- **Partial Observability:** The WEST path existed all along—we simply couldn't see it.

We formalize this as a *Semantic Turing Machine*: the tape is a sequence of natural language "pages," the head reads/writes scene descriptions, and the transition function is the causal dynamics of reality itself.

2.2 Kolmogorov Minimal Complexity Scenes

Each state is described by its Kolmogorov minimal complexity description—the shortest narrative that fully captures causally relevant information:

- **Raw:** "User log shows error 500 on /api/v1/auth. DB connection pool exhausted. Redis latency 200ms."
- **Minimal:** "Authentication failing due to DB connection exhaustion."

This compression aligns state representation with the "reasoning" space of LLMs—they process language, not tensors.

2.3 MCP Actions as Policy

The agent's action space is defined by the Model Context Protocol (MCP). The policy does not output arbitrary tokens; it calls tools:

- `observe(target)`: Gather information (restriction maps on the manifold).
- `interact(target, action)`: Modify state (traverse edges on the graph).
- `query(question)`: Ask the oracle/environment.
- `complete(status)`: Signal task completion.

2.4 Partial Observability and Belief Manifolds

The agent doesn't know the true state s_t—only an observation $o_t = \text{Render}(s_t)$. It maintains a belief distribution $b_t(s)$ over possible states.

We embed not just observations, but belief summaries:

```
Observation: "Dark forest. Paths N and E."
Belief: "I am in a forest. The lantern may be useful; forests often hide things."
```

High uncertainty spreads the embedding across the manifold; low uncertainty clusters it tightly. The Hodge Critic learns on this belief space.

2.5 Topological Ranking via Hodge Gradients

Instead of predicting a scalar reward, the critic assigns a topological gradient $\nabla\phi$ to each action. This gradient is the "clean" component of feedback after Hodge decomposition removes inconsistencies.

Actions are ranked by alignment with this gradient:

$$\text{score}(a) = \langle \text{embed}(a), \nabla\phi \rangle \tag{1}$$

Higher alignment = action moves along the geodesic toward good outcomes.

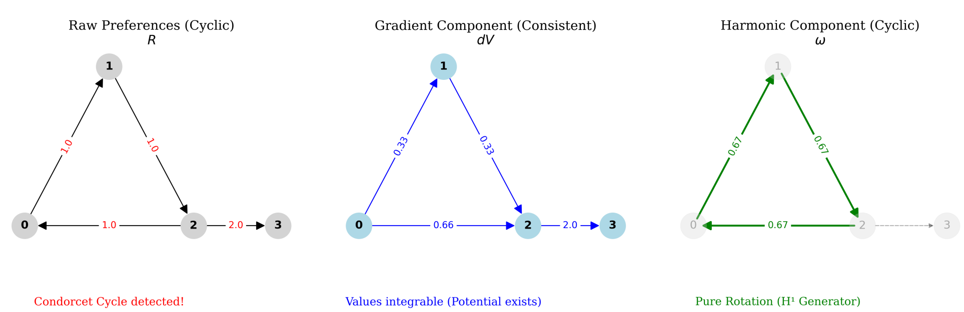

Figure 2: Hodge decomposition separates preference flow into gradient (consistent), curl (cyclic), and harmonic components. The gradient component defines the "true" reward direction.

3 Fine-Tuning Small LLMs as Agents

We target small, efficient models for tractable experimentation:

- Llama-3.2-3B-Instruct / Llama-3.1-8B-Instruct
- Mistral-7B-Instruct-v0.3
- Phi-3-mini-4k-instruct (3.8B)
- DeepSeek-Coder-V2-Lite-Instruct

3.1 Phase 1: Learning to Play

Supervised fine-tuning on (scene, action) pairs from:

- Text adventure transcripts (Jericho, TextWorld)
- Synthetic scenarios from GPT-4o oracle

3.2 Phase 2: Learning the Manifold (Geodesic DPO)

Instead of standard DPO loss, we use Geodesic DPO:

$$\mathcal{L}_{\text{GeoDPO}} = -\log \sigma(\beta \cdot \langle \nabla\phi, \Delta e \rangle) \tag{2}$$

Where $\nabla\phi$ is the Hodge gradient and Δe is the embedding difference between preferred/dispreferred actions. This aligns the policy with the manifold's gradient flow.

3.3 Phase 3: Inverse RL on the Manifold

The trained model's behavior defines an implicit reward:

$$R_{\text{implied}}(s, a) \propto \log \pi_\theta(a|s) \tag{3}$$

By embedding preferred actions across states, we reconstruct the reward manifold the model has internalized.

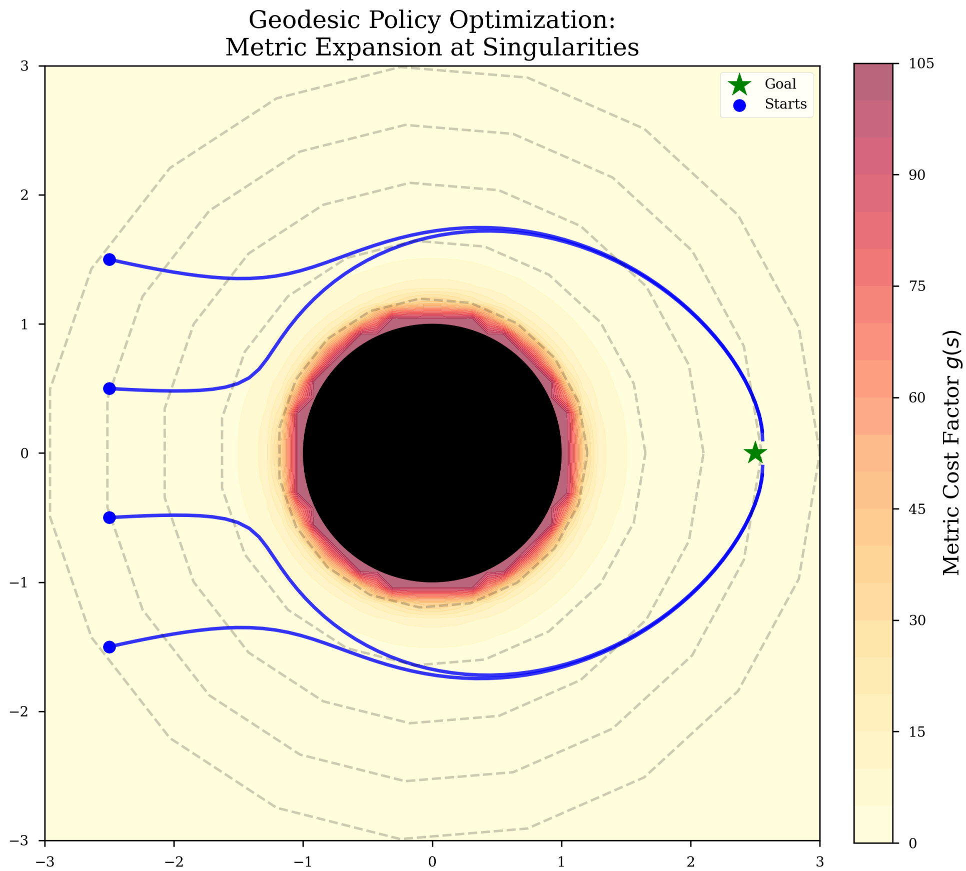

Figure 3: Geometric safety: "Black hole" regions represent unrecoverable unsafe states. The Riemannian metric creates infinite energy barriers preventing trajectories from entering these regions.

4 The Embedding Advantage

The key insight enabling this framework: *Natural language sequences are a universally applicable, but imperfect, state space that allows an interface for information that is trivially embeddable.*

Representation	**Embedding Difficulty**	**Semantic Richness**
Raw pixels	Hard (requires CNN)	Low
Structured JSON	Medium	Medium
Natural language	Easy (off-the-shelf)	High

Table 1: Comparison of state representations for embedding-based RL.

A single `sentence-transformers` model embeds:

- "The server is on fire" $\rightarrow$ Safety cliff region
- "User is satisfied" $\rightarrow$ High reward plateau
- "Contradictory requirements" $\rightarrow$ Topological defect ($H^1 \neq 0$)

Projection to 3D (via PCA/UMAP) makes the manifold visualizable and interpretable.

5 Conclusion & Summary

We are moving from "Reward Maximization" to "Manifold Navigation." By visualizing the reward space, stitching it together with Hodge theory, and grounding it in semantic state descriptions, we create a robust, interpretable, and mathematically rigorous framework for AI alignment.

In the research that follows, we will show experimental results in a reproducible RL environment demonstrating scalable improvements in reward hacking and other misalignment scenarios. We will then apply our Hodge critic and cohomology detection methods on a frontier-scale RLHF dataset to identify semantic inconsistencies and potential conflicts in the data for training safer and more predictable general language sequence agents.

Finally, we'll hypothesize on the success of PPO (i.e., clipped trust region policy-gradient approximation methods) and propose our framework as a more theoretically complete version of said methods (with caveats), wrapping up with key areas for improvement and research directions.